Nada Yahya Mousa

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Nada_LinkedIn | Nada_GitHub

EDUCATION

Software Engineering Diploma, ALX Africa Academy — Oct '24 — April '25
Technologies**:** Python, Databases(MYSQL), Django, Version control(Git/GitHub), Control flow, RESTful APIs, JWT, Linux, Postman.

Web Development Diploma, Let's know training center, Mansoura — Jun '19 — Sep '19
Technologies**:** JavaScript(ES6), TypeScript, HTML5,CSS3, Bootstrap, Sass / SCSS, Responsive Design, CI/CD for Frontend Builds,

Bachelor's degree in engineering, Damietta University — Jul '18 — Aug '23
Sector: Electrical and Electronics Engineering. Graduation Project: Control system design and modeling for a multi-terminal MMC-based HVDC grid for large renewable sources integration.

EXPERIENCE

Banque Misr — Jul '22 — Aug '22
Software Developer Intern — New Cairo, Egypt · (Remote)

- Experienced in front-end web development (Angular, TypeScript, REST APIs) with a strong background in systems design, and cross-functional technical collaboration. Built and launched an e-commerce web app[GitHub] .

Elsewedy Engineering Industries Company — Nov '23 — Present
Electrical Tendering Engineer — Al Jizah, Egypt · (On-site)

- Experienced in system design, technical communication, ERP (SAP, Odoo) workflows, and cross-functional project collaboration across automation, infrastructure, and industrial sectors.

SKILLS

Programming Languages:	Java, JavaScript, Python, C#, MATLAB/C, PLC
Software Technologies:	ASP.NET Core, Django, React.js, Bootstrap, MUI.
Databases & Backend:	MySQL, PostgreSQL, MongoDB RESTFUL API Development, ORM Tools.
DevOps and Tools	Version Control (Git/GitHub), Linux (Ubuntu), Vim, VSCode, Postman, Agile Methodologies.
Development Practices:	Object-Oriented Programming (OOP), SOLID Principles, Software Design Patterns.

PROJECTS

Data Migration Tool| [GitHub Link **]**
Technologies: Java**,** MongoDB, PostgreSQL , Cron, JSON.

- Build a data migration tool to sync data between MongoDB (source) and PostgreSQL (destination), using JSON mappings, cron scheduling, batch processing, Error handling.

E-Commerce (Artiqo Project) | [GitHub Link **]**
Technologies: Python (Django)**,** MySQL, RESTful APIs (JWT)/Postman.

- Designed and implemented secure, RESTful APIs for user authentication (JWT), product management (CRUD), cart operations, order processing, focusing on backend efficiency and data security.

Payment Application | [GitHub Link **]**
Technologies: C language , Software Design Patterns

- I implemented the SALE transaction simulating card, ATM terminal, and server interactions. Features include SALE (purchase payment), REFUND (return and refund), Pre-Authorization (temporary fund hold), and VOID (transaction cancellation).

On demand Traffic light control | [GitHub Link **]**
Technologies: C language, Proteus, Microchip Studio, Embedded system basics.

- Designed a real-time traffic light system with pedestrian crosswalk integration using AVR ATmega32 microcontroller and programmed it using C language. Tested the system in Proteus, ensuring reliable operation.

Electric Vehicle Prototype – Software & Systems Lead [GitHub link - website page**]**
DU Racing Team, Damietta University | Aug 2021 – Aug 2023 | 1st Place – EVER Egypt EV Competition (2021)

- Led embedded software and control system development for EV electrical architecture. Developed the team website.

Published Research Paper "Optimized Control of Modular Multilevel VSC-HVDC for Offshore Wind Farm Integration" __ MEPCON 2023 [Paper link]: Developed and optimized advanced control strategies for MMC-based HVDC systems integrating offshore wind farms, enhancing efficiency, reliability, and cost-effectiveness.